Reg. No. | | | | | | | | | | |

[14CE34]

THIRD SEMESTER B.TECH. EXAMINATION, DECEMBER –2015

SURVEYING - I

Time: 3 Hours **Maximum Marks:70**

Answer any FIVE full questions choosing at least two from each part.

PART - A

1. a. Explain the advantages of surveying in different phases of engineering. **04**
   b. Explain the basic principle of surveying. **04**
   c. Discus briefly the topographic maps, their numbering and scales. **06**

2. a. With a neat sketch describes the concept of "Reciprocal Ranging". **04**
   b. Explain the Hypotenusal allowance. **04**
   c. The distance between two villages measured by 20m chain was 1701m and when measured by a 30m chain it was 150 links. The test shows that both the chains were incorrect. What correction is required in the 20m chain, if the 30m chain is 0.4 links too long. **06**

3. a. How do you adjust the closing error graphically by Bowditch's Method? **07**
   b. The following are the beatings of a closed traverse ABCD at what station do you suspect local attraction. Find the correct Bearing of the sides if the Magnetic declination at the place is $2^0 20' W$. Compute the true bearing of the line. **08**

Line	AB	BC	CD	DA
FB	74°20'	$107^0 20'$	$224^0 50'$	$306^0 40'$
BB	$256^0 0'$	$286^0 20'$	$44^0 50'$	$126^0 0'$

4. a. Explain the principle of plane table surveying. **02**
   b. With a neat sketch explain 2 points problem. **08**
   c. Explain "strength of fix" in plane tabling. **04**

PART - B

5. a. Define Bench Mark? Explain the different types of Bench marks. **04**
   b. Explain the temporary adjustment of Dumpy level. **04**
   c. Explain Two-Peg test. **06**

6. a. Define the terms. **04**
   i. level surface
   ii. Height of collimination
   iii. reduced level
   iv. back sight

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b. Distinguish between longitudinal section and cross- section. **02**

c. **08**

Stations	B.S	I.S	F.S	Rise	Fall	RL	Remarks
1	2.150					450.00	BM-I
2	1.645		?	0.500		?	
3		2.345			?	?	
4	?		1.965	?		450.180	
5	2.050		1.825		0.400	449.78	
6	?		?	?		451.500	BM-II
7	1.690		1.570	0.120		451.620	
8	2.865		2.100		?	451.210	
			?	?		451.250	BM-III

A page of level Book is shown in the following table.

Fill the missing readings and calculate the RL of all the points and apply the check.

7. a. Explain the characteristics of contour. **04**

b. Sketch contours to represents the following. **04**

i. Uniform Slope

ii. Valley

iii. Ridge

iv. Vertical Cliff

c. The following perpendicular offsets were taken at 10m intervals from a survey line to an irregular boundary line: **06**

3.25, 5.60, 4.20, 6.65, 8.75, 6.20, 3.25, 4.20, 5.65.

Calculate the area enclosed between the survey line, the irregular boundary line and the first and last offsets by the application of

i. Trapezoidal rule

ii. Simpson's rule.

8. a. Explain the fundamental lines and their desired relation of a transit theodolite. **05**

b. What is "double transit method"? Describe the extension of a straight line in the field by this method? **05**

c. Write a note on Total station and list the fundamental measurements made by it. **04**